Smart Factory Equipment Energy Consumption Prediction

Abstract

This report presents a descriptive study on forecasting equipment energy consumption in a smart manufacturing facility using sensor network data. We outline data characteristics, exploratory analysis, feature selection, baseline and tuned model development, and performance evaluation. The goal is to provide actionable insights for energy optimization.

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1. Introduction

Rising energy costs in manufacturing drive the need for predictive systems to forecast energy consumption and enable proactive efficiency measures. SmartManufacture Inc. supplied a dataset of environmental and operational sensor readings across nine factory zones, along with external weather conditions, to develop regression models for predicting equipment energy usage (`equipment_energy_consumption`). This report synthesizes the end-to-end workflow and findings from the `energy_prediction.ipynb`.

2. Data Description

The dataset comprises 15,880 records split into training (11,116), validation (2,382), and testing (2,382) sets. Each record includes:

- Timestamps (YYYY-MM-DD HH:MM:SS)
- Energy metrics: `equipment_energy_consumption`, `lighting_energy`
- Zone measurements: temperature & humidity for Zones 1–9
- External weather: `wind_speed`, `outdoor_temperature`, `outdoor_humidity`, `atmospheric_pressure`,, `visibility_index`, `dew_point`
- Random variables: `random_variable1`, `random_variable2` (to be evaluated for inclusion)

Detailed feature descriptions are provided in the data description document.

3. Exploratory Data Analysis

- Distribution of target: Right-skewed energy consumption, mostly in lower Wh range.
- Temporal patterns: Hour-of-day and day-of-week effects on energy draw align with production cycles.

- Correlation analysis identified strong relationships between equipment consumption and environmental factors such as zone temperatures, outdoor humidity, and atmospheric pressure.

- Outliers & missing values: Minimal missing data; outliers capped at the 1st and 99th percentiles.

Key visualizations (histograms, box plots, correlation heatmap) were generated to support these insights.

4. Feature Selection

A LightGBM-based importance ranking showed that features contributing less than 1.5% importance were dropped: `hour`, `month_of_year`, `day_of_week`. Remaining predictors include lighting energy, zone measurements, outdoor weather, and dew point. Top 5 features by importance:

Feature	Importance Score
dewp_point	159
zone5_humidity	148
zone3_temperature	137
hour_of_day	136
zone2_temperature	131

(Full importance list available in the notebook.)

5. Model Development and Evaluation

We implemented six baseline regressors on the validation set:

Model	RMSE	MAE	R^2
LightGBM	167.67	70.29	0.1000
Random Forest	169.90	72.29	0.0759
Gradient Boosting	170.54	72.03	0.0690
Ridge Regression	174.85	77.02	0.0212
Linear Regression	174.85	77.02	0.0212
XGBoost	176.70	84.55	0.0005

LightGBM achieved the best baseline performance (RMSE: 167.67, MAE: 70.29, R^2: 0.1000).

6. Hyperparameter Tuning

A randomized search over 50 parameter combinations for LightGBM with 5-fold cross-validation was conducted using 5-fold cross-validation on the training set. The search space included:

- `num_leaves`: 20–150
- `max_depth`: 3–12
- `learning_rate`: 0.01–0.20
- `n_estimators`: 100–1000
- `min_child_samples`: 20–100
- `subsample`, `colsample_bytree`: 0.6–1.0
- `reg_alpha`, `reg_lambda`: 0.0–1.0

Best parameters found:

colsample_bytree: 0.8476

learning_rate: 0.0251

max_depth: 10

min_child_samples: 67

n_estimators: 354

num_leaves: 23

reg_alpha: 0.1075

reg_lambda: 0.6412

subsample: 0.7074

Tuned LightGBM performance on validation set: RMSE: 167.300, MAE: 68.107, R^2: 0.104.

7. Test Set Evaluation

The tuned LightGBM model was evaluated on the independent test set:

- RMSE: 174.931
- MAE: 68.845
- R^2: 0.081

Performance degradation reflects distributional differences between validation and test splits, suggesting potential for further feature or model refinement.

8. Discussion

- Model accuracy remains modest (R^2 ≈0.08–0.10), indicating that additional predictors (e.g., operational schedules, equipment load data) may be needed.
- Feature influence: Environmental factors within the factory zones and lighting energy are key drivers.
- Temporal features: Dropped due to low importance, but cyclic patterns suggest refined encoding (e.g., sin/cos transforms) could improve performance.

9. Conclusion and Recommendations

- Implementation: Deploy the tuned LightGBM model in a monitoring dashboard for real-time energy forecasts.
- Data enrichment: Integrate operational metadata (e.g., shift schedules, equipment utilization) to enhance predictive power.
- Model improvement: Explore advanced time-series models (LSTM, Prophet) and ensemble stacking.
- Energy management: Use forecasts to schedule high-energy tasks during off-peak hours and adjust environmental controls.

References

1. Smart Factory Data Description Document
2. Energy Prediction Notebook (`energy_prediction.ipynb`)